
Sourav Paul, Ph.D.

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EMPLOYMENT

2025 - Present	University of Toronto, Toronto, Canada Postdoctoral Fellow, Department of Materials Science and Engineering Advisor: Prof. Chandra Veer Singh
2023 - 2024	University of Alberta, Edmonton, Canada Overseas Visiting Doctoral Fellow, Department of Chemical and Materials Engineering Advisor: Prof. Samir H. Mushrif
2020 - 2025	Ramakrishna Mission Vidyamandira, University of Calcutta, India Research Fellow, Department of Industrial Chemistry and Applied Chemistry Advisor: Dr. Uttam Kumar Ghorai

EDUCATION

2020 - 2025	University of Calcutta, India Ph.D. in Applied Chemistry Thesis: Electrochemical fixation of N ₂ and CO ₂ using cobalt phthalocyanine based catalysts
2017 - 2019	Indian Institute of Technology Kharagpur, India M.Tech. in Rubber Technology Thesis: Evaluation of graphene as a multifunctional nanofiller in rubber compounds
2015 - 2017	University of Calcutta, India M.Sc. in Applied Chemistry Thesis: Development of copper phthalocyanine nanostructures by thermal evaporation
2012 - 2015	University of Calcutta, India B.Sc. in Industrial Chemistry Ramakrishna Mission Vidyamandira

RESEARCH INTERESTS

Computational materials science; density functional theory; molecular dynamics; multiscale modelling; electrocatalysis; electrochemical reaction engineering; carbon dioxide capture and conversion; green ammonia, urea, and nitric acid synthesis; two-dimensional materials; machine learning assisted catalyst discovery.

SELECTED RESEARCH CONTRIBUTIONS

- Engineered cobalt phthalocyanine nanotubes and 1D/2D heterostructures for nitrogen reduction, nitrate reduction, nitrogen oxidation, and electrochemical carbon-nitrogen coupling.
- Integrated DFT, charge analysis, molecular dynamics, electrochemical experiments, and COMSOL simulations to connect active-site chemistry, interfacial electric fields, and catalytic performance.
- Developed mechanistic and materials-design frameworks for renewable-electricity-driven synthesis of ammonia, urea, and nitric acid under ambient conditions.
- Co-inventor on two United States and three Indian patent applications covering electrochemical synthesis of ammonia, green urea, and nitric acid.

HONORS AND AWARDS

2024	Finalist, Best Paper Award in Carbon Capture and Utilization, ICCCU, JNCASR
2024	Certificate of Patent Award for patent development and commercial application, Ramakrishna Mission Vidyamandira
2023	Overseas Visiting Doctoral Fellowship, SERB, Government of India
2023	Best Poster Award, Sustainable Functional Materials and Processes, IIT Kharagpur
2023	Emerging Scholar recognition, ACS Sustainable Chemistry and Engineering
2021	Best Poster Award, Chemical Science Symposium, Royal Society of Chemistry
2021	Acharya Prafulla Chandra Ray Research Excellence, Indian Chemical Society
2017	GATE All India Rank 63 and UGC Postgraduate Merit Scholarship (2 nd rank holder in University [Chemical Sciences Category])
2016	Summer Research Fellowship, Indian Academy of Sciences

PUBLICATIONS

21 peer-reviewed journal publications | Google Scholar profile: [view publications](#)

1. **S. Paul, P. S. Maiti, K. Sahoo, B. Keshari, T. K. Dhar, U. K. Ghorai.** Emerging 2D Materials for Green Ammonia Electrosynthesis: Integrating Experimental, Computation, and Machine Learning Strategies Toward Commercialization. *ChemSusChem* 19, e202502111 (2026). DOI: 10.1002/cssc.202502111.
2. **S. Paul, A. Gain, A. Adalder, S. Pal, S. H. Mushrif, U. K. Ghorai.** Morphology Dependent N₂ Reduction on Cu₂O: Combined Experimental and Computational Study for Efficient Ammonia Electrocatalytic Synthesis. *Small*, 2505121 (2025). DOI: 10.1002/smll.202505121.
3. **S. Paul, A. Adalder, N. Barman, R. Thapa, A. Bera, K. Mitra, U. K. Ghorai.** Boosting Selective Nitrogen Oxidation to Nitric Acid by Synergizing Cobalt Phthalocyanine on Carbon Nitride Surface. *Advanced Functional Materials* 34, 2408314 (2024). DOI: 10.1002/adfm.202408314.
4. **S. Paul, S. Sarkar, A. Adalder, A. Banerjee, U. K. Ghorai.** Dual Metal Site-Mediated Efficient C-N Coupling toward Electrochemical Urea Synthesis. *Journal of Materials Chemistry A* 11, 13249-13254 (2023). DOI: 10.1039/D3TA01011B.
5. **S. Paul, S. Sarkar, A. Adalder, S. Kapse, R. Thapa, U. K. Ghorai.** Strengthening the Metal Center of Co-N₄ Active Sites in a 1D-2D Heterostructure for Nitrate and Nitrogen Reduction Reaction to Ammonia. *ACS Sustainable Chemistry and Engineering* 11, 6191-6200 (2023). DOI: 10.1021/acssuschemeng.2c07114.
6. **S. Paul, S. Sarkar, et al..** 1D/2D Interface Engineering of a CoPc-C₃N₄ Heterostructure for Boosting the Nitrogen Reduction Reaction to NH₃. *Dalton Transactions* 52, 15360-15364 (2023). DOI: 10.1039/D3DT01790G.
7. **S. Paul, A. Adalder, U. K. Ghorai.** Progress of Electrocatalytic Urea Synthesis: Strategic Design, Reactor Engineering, Mechanistic Details and Techno-Commercial Study. *Materials Chemistry Frontiers* 7, 3820-3854 (2023). DOI: 10.1039/D3QM00433C.
8. **S. Paul, A. Adalder, et al..** Alq₃-Graphene Bifunctional Nanocomposite for Cold Cathode and Luminescent Anode in Field Emission Display Applications. *Journal of Electronic Materials* (2022). DOI: 10.1007/s11664-022-10009-x.
9. **U. K. Ghorai, S. Paul, B. Ghorai, A. Adalder, S. Kapse, R. Thapa, A. Nagendra, A. Gain.** Scalable Production of Cobalt Phthalocyanine Nanotubes: Efficient and Robust Hollow Electrocatalyst for Ammonia Synthesis at Room Temperature. *ACS Nano* 15, 5230-5239 (2021). DOI: 10.1021/acsnano.0c10596.
10. **J. Mukherjee, S. Paul, A. Adalder, S. Kapse, R. Thapa, S. Mandal, B. Ghorai, S. Sarkar, U. K. Ghorai.** Understanding the Site-Selective Electrocatalytic Co-Reduction Mechanism for Green Urea Synthesis Using Copper Phthalocyanine Nanotubes. *Advanced Functional Materials* 32, 2200882 (2022). DOI: 10.1002/adfm.202200882.

Complete publication list available through the embedded Google Scholar link.

PATENTS

- Process for the Electrochemical Synthesis of Green Urea, United States patent application 17846753, filed 2022.
- Process for the Electrochemical Synthesis of Ammonia, United States patent application, filed 2021.
- Indian patent applications 202131029798, 202131027290, and 202131029797 covering ammonia, green urea, and nitric acid synthesis, filed 2021.

TEACHING EXPERIENCE AND MENTORSHIP

2020 - 2023

Ramakrishna Mission Vidyamandira, India

Teaching Assistant

General Chemistry laboratories and scientific presentations; mentored three undergraduate and two masters students, contributing to one publication and two conference presentations.

2023 - 2024

University of Alberta, Canada

Research Mentor

Guided an international undergraduate exchange student on DFT-based electrocatalyst design for acetamide synthesis.

TECHNICAL EXPERTISE

- Computational: VASP, DFT and DFT+U, GROMACS, molecular dynamics, COMSOL Multiphysics, VMD, Bader charge and electronic-structure analysis.
- Electrochemical and materials: catalyst synthesis, electrochemical workstation, rotating electrodes, XRD, XPS, XANES, Raman, FTIR, UV-Vis, BET, SEM, TEM, TGA, DSC, and DMA.
- Analysis: Artemis, Athena, Hephaestus, CASA, Origin Pro; mechanistic interpretation, structure-property analysis, and multiscale model development.

ADDITIONAL RESEARCH EXPERIENCE

2018 - 2019

Indian Institute of Technology Kharagpur and Birla Carbon, India

M.Tech. Thesis Researcher: Graphene-like materials in polar and nonpolar rubber systems; dispersion optimization; mechanical, electrical, and tire-performance analysis.

2017

Ramakrishna Mission Vidyamandira, India

M.Sc. Thesis Researcher: Thermal vapor deposition of copper phthalocyanine on rigid and flexible substrates; XPS, SEM, TEM, photoluminescence, and cathodoluminescence studies.

2016

Institute of Nanoscience and Technology, Mohali, India

Summer Research Fellow: Silicon quantum dots prepared through hydrothermal and microfluidic routes; photoluminescence analysis.

2015

Saha Institute of Nuclear Physics, India

Undergraduate Researcher: Spectroscopic investigation of bovine serum albumin interactions with gold nanoparticles and riboflavin-conjugated gold nanoparticles.

PROFESSIONAL REFERENCES

Prof. Chandra Veer Singh

Postdoctoral Advisor | Professor, Department of Materials Science and Engineering, University of Toronto
chandraveer.singh@utoronto.ca

Dr. Uttam Kumar Ghorai

Ph.D. Advisor | Assistant Professor and Head, Department of Industrial Chemistry and Applied Chemistry, Ramakrishna Mission Vidyamandira
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Prof. Samir H. Mushrif

Overseas Doctoral Research Advisor | Professor, Department of Chemical and Materials Engineering, University of Alberta
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Prof. Anil K. Bhowmick

M.Tech. Thesis Advisor | Research Professor, Department of Chemical and Biomolecular Engineering, University of Houston
bhowmick@central.uh.edu